COMPUTER NETWORKS (MIC23) — UNIT III

MAC Sublayer & LANs — Handwritten Study Notes

Student Blue-Pen Notebook Style • Short & Direct 2M Answers • Complete Visual Diagrams for Every Question • All Source Papers

This Master Question & Answer document combines all exam questions from:

- **Model Paper-1** (MIC23 B.Tech CSE/AIML)
- **Model Paper-2** (MIC23 B.Tech CSE/AIML)
- **Previous University Exam Paper** (April-2026 Regular/Supple)
- **Mid-1 & Mid-2 Important Questions**

All answers are rewritten in **extremely simple English** that any beginner student can read once and understand immediately. Every question includes a relevant visual diagram, real-life analogies, box comparison tables, step-by-step numerical derivations, and full examination structures.

SECTION A: 2-MARK QUESTIONS & ANSWERS

Short, direct answers (~40–65 words) written in simple English with student pen-style diagrams

(Short, direct answers suitable for 2-mark university exams. Target: 40–65 words)

Q1. *(Paper-1)*

Briefly Explain the core difference between random access and controlled access protocols?

Definition:

- **Random Access:** Any computer can transmit whenever it has data to send without asking for permission.
- **Controlled Access:** Computers can only transmit after receiving permission or a token from a controller.

Feature	Random Access	Controlled Access
Permission	No permission needed; transmit freely.	Must wait for turn, token, or poll.
Data Collisions	High chance of packet crashes.	Zero data collisions guaranteed.
Examples	ALOHA, CSMA/CD, CSMA/CA.	Reservation, Polling, Token Ring.

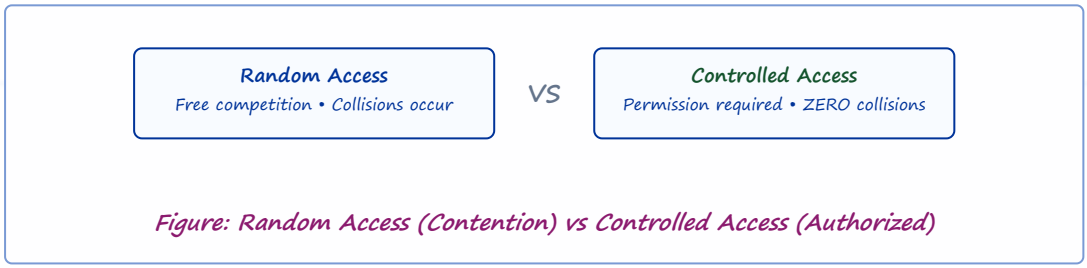


Figure: Random Access (Contention) vs Controlled Access (Authorized)

Q2. *(Paper-1)*

Describe the working of pure ALOHA protocol?

Definition:

Pure ALOHA is the simplest random access protocol where a computer transmits its data frame immediately whenever it has data ready.

Key Points:

- **Continuous Time:** There are no time slots; stations transmit at any random instant.
- **High Collision Rate:** If another station transmits even one bit at the same time, a data crash occurs.
- **Maximum Efficiency:** Maximum throughput is very low, only **18.4%** (1/2e).

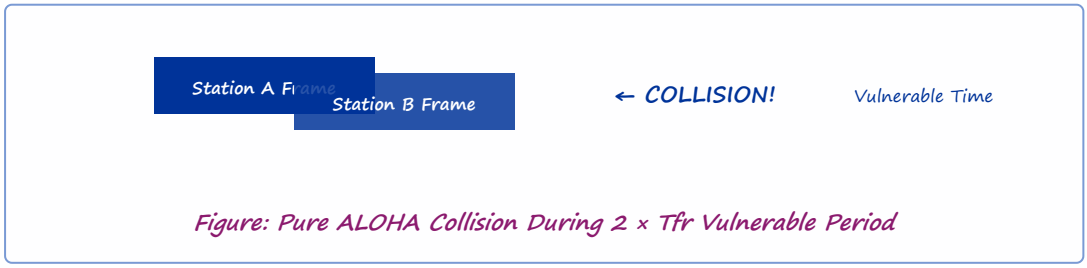


Figure: Pure ALOHA Collision During 2 × Tfr Vulnerable Period

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Q3. *(Paper-2)*

What is CSMA and how does it reduce collisions?

Definition:

CSMA (Carrier Sense Multiple Access) is a random access protocol based on the principle *"Listen Before You Talk".*

How It Reduces Collisions:

- **Sensing the Cable:** A station checks whether the shared wire is idle or busy before sending data.
- **Waits If Busy:** If another station is already transmitting, the station waits, preventing unnecessary packet crashes.



Figure: CSMA "Listen Before Speaking" Decision Logic

Q4. *(Paper-2)*

Compare FDMA, TDMA, and CDMA in channelization?

Definition:

Channelization divides a single shared cable or radio band among multiple users so they do not interfere with each other.

Technique	Dividing Factor	How Users Share
FDMA	Frequency	Entire time, but uses a private narrow frequency band.
TDMA	Time	Entire frequency band, but uses a private allocated time slot.
CDMA	Math Code	Uses entire frequency band all the time using unique orthogonal codes.



Figure: Three Core Channelization Dimensions (Frequency • Time • Code)

Q5. *(Paper-3)*

What is random access in MAC protocols?

Definition:

Random Access is a method used in the Medium Access Control (MAC) sublayer where multiple computers share one common channel without any central master controller.

Key Points:

- **No Fixed Turns:** No computer has a fixed schedule or priority over others.
- **Collision Handling:** If two stations transmit at once, a collision occurs; both wait a random time and retransmit.

Q6. *(Paper-3)*

Define Carrier Sense Multiple Access (CSMA).

Definition:

CSMA is a random access MAC protocol in which a computer listens to the communication channel to check if it is free before starting transmission.

Key Points:

- If channel is **free**, the station transmits its frame.
- If channel is **busy**, the station waits using 1-persistent, non-persistent, or p-persistent strategies to reduce data crashes.

Q7. *(MID-1 / MID-2)*

Differentiate between CSMA/CD and CSMA/CA.

Definition:

- **CSMA/CD:** Detects data crashes *after* they happen (used in wired Ethernet).
- **CSMA/CA:** Tries to avoid data crashes *before* transmitting (used in wireless Wi-Fi).

Feature	CSMA/CD (Collision Detection)	CSMA/CA (Collision Avoidance)
Action	Detects collisions while transmitting.	Prevents collisions before transmitting.
On Collision	Aborts immediately and sends jam signal.	Uses waiting time (IFS) and RTS/CTS signals.
Medium	Traditional wired Ethernet cables.	Wireless networks (Wi-Fi, 802.11).

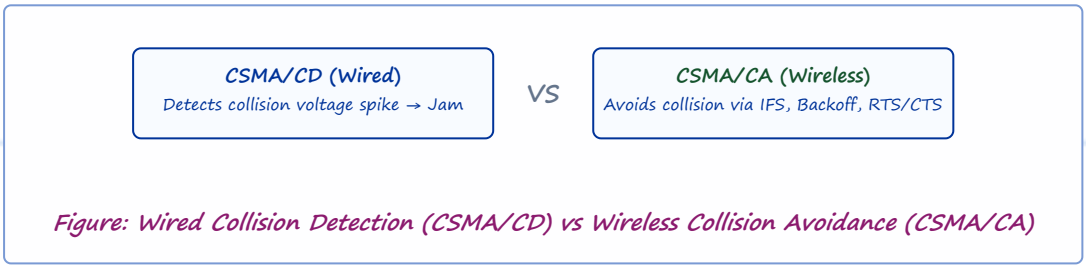


Figure: Wired Collision Detection (CSMA/CD) vs Wireless Collision Avoidance (CSMA/CA)

Q8. *(MID-1 / MID-2)*

What is channelization? Name the channelization techniques.

Definition:

Channelization is a method of dividing one shared physical communication channel among multiple users without signals mixing or interfering.

Three Main Techniques:

- **FDMA (Frequency Division Multiple Access):** Splits frequency band into smaller channels.
- **TDMA (Time Division Multiple Access):** Splits time into discrete transmission slots.
- **CDMA (Code Division Multiple Access):** Assigns unique mathematical codes to each user.

Q9. *(Paper-3)*

List the services provided by the network layer to the transport layer.

Definition:

The **Network Layer (Layer 3)** moves packets from the source computer to the destination computer across multiple networks on behalf of the Transport Layer.

Key Services:

- **Packetizing:** Wraps transport layer segments into network packets with IP headers.
- **Routing:** Finds the shortest and most optimal route through multiple intermediate routers.
- **Logical Addressing:** Adds source and destination IP addresses so packets reach the correct machine.

SECTION B: 5-MARK QUESTIONS (FULL 10-MARK EXAM STRUCTURE)

Detailed answers with Introduction, Dual Diagrams (Standard Exam + Intuitive Analogy), Box Comparison Tables, and Step-by-Step Derivations

(All prepared using complete 10-Mark Answer Structure: Introduction, Definition, Detailed Points, Dual Diagrams, Step-by-Step Calculations, Real-Life Examples, Box Tables, and Conclusions)

Q1. *(Paper-1)*

Discuss about CSMA/CA protocol and CSMA/CD protocol?

1. Introduction

When computers share a communication medium, they must follow rules to prevent two devices from transmitting at the exact same time. The two most famous contention-based protocols are **CSMA/CD** (used in wired Ethernet) and **CSMA/CA** (used in wireless Wi-Fi).

2. Detailed Explanation: CSMA/CD (Collision Detection)

- **Used In:** Wired networks (Ethernet IEEE 802.3).
- **How It Works:**
  - **Sense:** Station listens to the wire. If idle, it transmits.
  - **Listen while Transmitting:** While sending data, the station continuously measures the electrical voltage on the wire.
  - **Detect:** If another station also transmitted, the two electrical voltages add together, creating a high voltage spike — a **Collision**!

• **Abort & Jam:** The station aborts transmission immediately, broadcasts a 32-bit **Jam Signal** to alert all other stations, and waits a random time calculated by the **Binary Exponential Backoff** algorithm before trying again.

3. Detailed Explanation: CSMA/CA (Collision Avoidance)

- **Used In:** Wireless networks (Wi-Fi IEEE 802.11).
- **Why Detection Doesn't Work in Wireless:** Wireless radios cannot transmit and listen at the same time because the outgoing antenna power completely drowns out weak incoming signals.
- **How Avoidance Works (The 3 Precautions):**

• **IFS (Inter-Frame Space):** When the channel becomes idle, the station does not transmit immediately. It waits for a mandatory quiet period called IFS.

• **Contention Window (Random Backoff):** The station picks a random number of time slots to wait. It only transmits if the channel remains completely silent.

• **RTS / CTS Handshake:**

- Sender sends a tiny **RTS (Request to Send)** frame.
- Receiver replies with a **CTS (Clear to Send)** frame.
- All other wireless devices hear the CTS and stay quiet for the duration of the transmission.

• **Acknowledgment (ACK):** The receiver sends an ACK. If no ACK arrives, a collision is assumed.

4. Real-Life Analogy

- **CSMA/CD is like People Talking in a Dark Room:** Everyone listens. If two people start speaking at the same second, they hear each other's voices collide, shout "Oops!" (Jam Signal), go silent, wait a few seconds, and then speak one at a time.
- **CSMA/CA is like a Formal Classroom:** Students do not just shout. You raise your hand (**RTS**). The teacher points at you and says "Go ahead, Rahul" (**CTS**). Everyone else stays quiet while Rahul speaks.

5. Dual Diagrams

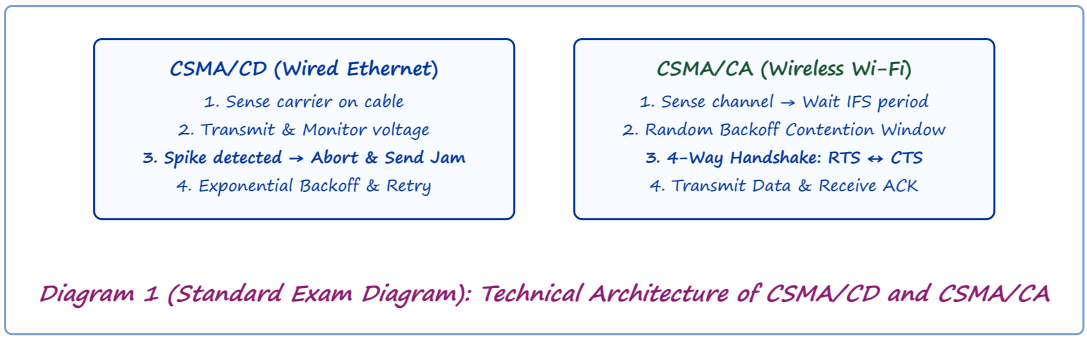


Diagram 1 (Standard Exam Diagram): Technical Architecture of CSMA/CD and CSMA/CA

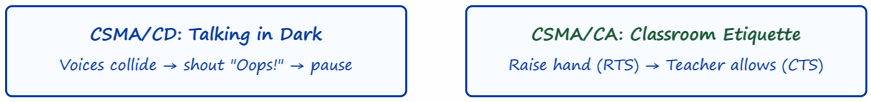


Diagram 2 (Easy Understanding Diagram): Everyday Analogies for Collision Detection vs Avoidance

Diagram 1: Standard Exam Diagram

Shows vertical flowchart of CSMA/CD (Sense → Transmit → Collision Detected? → Jam → Backoff) side-by-side with CSMA/CA (Sense → Wait IFS → Backoff → RTS/CTS → Data → ACK).

Diagram 2: Easy Understanding Diagram

Shows wired electrical voltage spike on cable (CSMA/CD) versus classroom student raising hand to speak (CSMA/CA).

6. Comprehensive Box Comparison: CSMA/CD vs CSMA/CA

Parameter	CSMA/CD	CSMA/CA
Full Form	Carrier Sense Multiple Access with **Collision Detection**.	Carrier Sense Multiple Access with **Collision Avoidance**.
Primary Environment	**Wired Networks** (Ethernet IEEE 802.3).	**Wireless Networks** (Wi-Fi IEEE 802.11).
Collision Strategy	Detects collisions **during** transmission.	Prevents collisions **before** transmission.
Action on Collision	Stops immediately and sends **Jam Signal**.	Cannot detect; infers collision when ACK is missing.
Control Frames	None.	Uses **RTS** (Request to Send) and **CTS** (Clear to Send).

Parameter	CSMA/CD	CSMA/CA
Bandwidth Overhead	Extremely low.	Moderate (due to IFS, backoff, and RTS/CTS).

7. Conclusion

CSMA/CD provides high efficiency on physical wires by quickly detecting voltage collisions and aborting, while CSMA/CA safely navigates the tricky wireless airwaves by reserving the channel before transmitting.

Q2. *(Paper-1)*

Explain different types of Controlled Access methods?

1. Introduction

In random access protocols, stations compete for the channel, leading to collisions when network traffic is heavy. **Controlled Access Protocols** eliminate collisions completely by enforcing strict rules on which station is authorized to transmit.

2. Definition

Controlled Access is a category of Medium Access Control (MAC) protocols where stations consult one another or a central controller to find out which station has the right to transmit. A station cannot transmit unless authorized.

3. Three Major Controlled Access Methods

A. Reservation Method

- Time is divided into intervals. Before each data frame interval, there is a small **Reservation Frame**.
- If there are N stations, the reservation frame has N small mini-slots.
- Station k marks a 1 in mini-slot k if it has data to send.
- After the reservation frame, stations transmit their data frames strictly in the order of their reservations.
- **Analogy:** Reserving a seat in a movie theater before the show starts.

B. Polling Method

- Works in a **Primary-Secondary (Master-Slave)** architecture.
- One central computer is the **Primary Station**; all other computers are **Secondary Stations**.

- **Poll Operation:** When the primary wants to receive data, it asks secondary stations one by one: *"Station A, do you have data?"* If Station A replies NAK (no), it asks Station B.
- **Select Operation:** When the primary wants to send data, it sends a `Select` message: *"Station A, are you ready to receive data?"* Once Station A replies ACK, the primary transmits.
- **Analogy:** A school teacher calling on students one by one to answer a question.

C. Token Passing Method

- Stations are organized in a logical ring.
- A small special frame called a **Token** circulates constantly from station to station.
- If a station wants to transmit, it catches the token, attaches its data, and sends it along the ring.
- While the station holds the token, no other station is allowed to transmit.
- When finished, the station releases the token for the next station.
- **Analogy:** A microphone passed around a boardroom meeting; only the person holding the microphone is permitted to speak.

4. Dual Diagrams

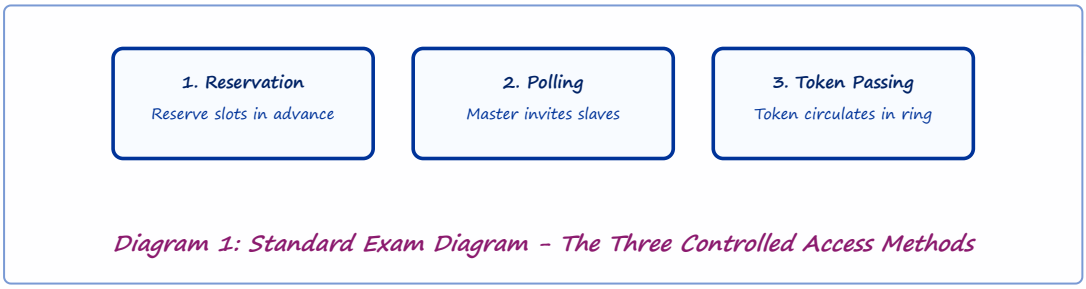


Diagram 1: Standard Exam Diagram – The Three Controlled Access Methods

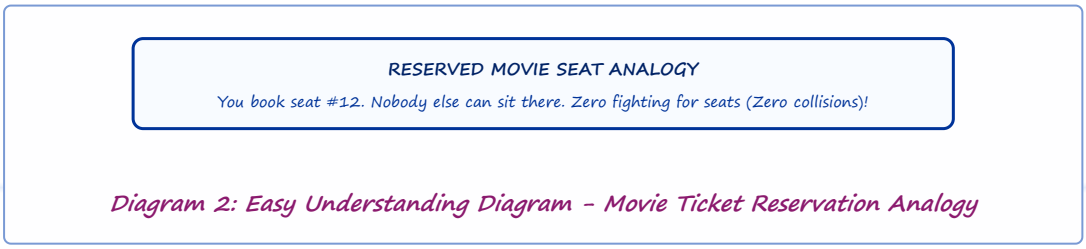


Diagram 2: Easy Understanding Diagram – Movie Ticket Reservation Analogy

Diagram 1: Standard Exam Diagram

Shows three panels:

- Reservation mini-slots preceding data frames.
- Primary Master polling Secondary Slaves.
- Logical Ring with Token passing between nodes.

Diagram 2: Easy Understanding Diagram

Shows theater seat reservation tickets, teacher pointing at students, and microphone passed in a circle.

5. Comparison Table of Controlled Access Methods

Parameter	Reservation Method	Polling Method	Token Passing Method
Architecture	Distributed / Peer-to-peer.	Centralized (Master-Slave).	Logical Ring.
Control Entity	Predefined reservation slots.	Central Primary Computer.	Circulating Token frame.
Failure Vulnerability	Low.	High (if Primary crashes, all stops).	High (token lost crashes ring unless recovered).
Collision Probability	**Zero**.	**Zero**.	**Zero**.

6. Conclusion

Controlled access methods eliminate collisions and ensure fair channel sharing, making them ideal for mission-critical networks where guaranteed delivery times are mandatory.

Q3. *(Paper-1)*

Explain Standard Ethernet, Fast Ethernet (100 Mbps)?

1. Introduction

Ethernet is the most widely installed Local Area Network (LAN) technology in the world. Originally developed by Xerox in the 1970s and standardized by IEEE as **802.3**, Ethernet has evolved from early 10 Mbps coax cables to high-speed gigabit optical links.

2. Detailed Explanation: Standard Ethernet (10 Mbps)

- **Speed:** 10 Mbps (Megabits per second).
- **Medium:** Thick coaxial cable (10BASE5), thin coaxial (10BASE2), or unshielded twisted pair (10BASE-T).
- **MAC Protocol: CSMA/CD** (Carrier Sense Multiple Access with Collision Detection).
- **Minimum Frame Size:** Exactly **64 Bytes (512 bits)**.

- *Why 64 Bytes?* A station must transmit for at least 2 imes ext{Propagation Delay} (2 au) so that if a collision occurs at the far end of the cable, the sender is still transmitting and can detect the voltage spike!
- **Maximum Frame Size: 1518 Bytes** (to prevent one computer from hogging the wire).

3. Detailed Explanation: Fast Ethernet (100 Mbps — IEEE 802.3u)

- **Speed: 100 Mbps** (10 times faster than Standard Ethernet!).
- **Backward Compatibility:** Uses the exact same 64-byte to 1518-byte frame format as Standard Ethernet so software does not need to change.
- **Distance Adjustment:** Because data travels 10 times faster, the transmission time of a 64-byte frame is 10 times shorter! To keep CSMA/CD working, the maximum network collision domain radius had to be reduced from 2500 meters down to **250 meters** (or 100 meters per twisted-pair segment).
- **Media Standards:**
- **100BASE-TX:** Uses 2 pairs of Category 5 UTP copper wire (most popular).
- **100BASE-FX:** Uses 2 strands of optical fiber for distances up to 2 km.
- **Autonegotiation:** Network cards automatically detect if the other end is 10 Mbps or 100 Mbps and switch speeds accordingly.

4. Dual Diagrams



Diagram 1 (Standard Exam Diagram): IEEE 802.3 Standard Ethernet Frame Structure



Diagram 2 (Easy Understanding Diagram): Standard vs Fast Ethernet Speed Multiplier

Diagram 1: Standard Exam Diagram

Shows the IEEE 802.3 Ethernet Frame format: Preamble (7B) • SFD (1B) • Dest MAC (6B) • Source MAC (6B) • Length/Type (2B) • Data (46-1500B) • CRC FCS (4B).

Diagram 2: Easy Understanding Diagram

Shows 10 Mbps single-lane road compared with 100 Mbps multi-lane expressway carrying identical standard shipping containers (Ethernet frames).

5. Comparison Table: Standard Ethernet vs Fast Ethernet

Parameter	Standard Ethernet (IEEE 802.3)	Fast Ethernet (IEEE 802.3u)
Data Rate	**10 Mbps**	**100 Mbps**
Bit Duration	100 ext{ ns}	10 ext{ ns} (10 times narrower bits).
Cable Types	Coaxial (10BASE5/2), Cat3 UTP.	Cat5 UTP (100BASE-TX), Fiber (100BASE-FX).
Maximum Network Span	2500 ext{ meters}	250 ext{ meters} (100m on UTP).
Minimum Frame Size	64 Bytes (512 ext{ bits}).	64 Bytes (512 ext{ bits} — identical!).

6. Conclusion

Fast Ethernet preserved the universal 802.3 frame structure while multiplying throughput by 10, establishing Ethernet as the permanent standard for office and enterprise LANs.

Q4. *(Paper-1)*

Explain Frequency Division Multiple Access (FDMA)?

1. Introduction

When multiple radio transmitters share the same open airspace, their wireless signals collide and become garbled unless separated. **FDMA (Frequency Division Multiple Access)** is the oldest and most natural way to share airspace, used in broadcast radio and 1G analog cellular systems.

2. Definition & Core Working

- **FDMA:** A channelization access method where the total available frequency bandwidth is divided into distinct, non-overlapping frequency bands (sub-channels).
- Each station is assigned its own private frequency band.
- All stations can transmit **at the exact same time**, but on **different frequencies**.

- **Guard Bands:** Narrow unused frequency gaps placed between adjacent channels to prevent signals from overlapping and causing interference.

3. Real-Life Analogy

Think of **FM Radio Broadcasting**:

- All radio stations broadcast their music at the exact same second into the city air.
- Radio Mirchi broadcasts at 98.3 MHz.
- Red FM broadcasts at 93.5 MHz.
- Your radio car tuner tunes into 98.3 MHz and hears only Radio Mirchi cleanly because the frequencies never touch.

4. Dual Diagrams

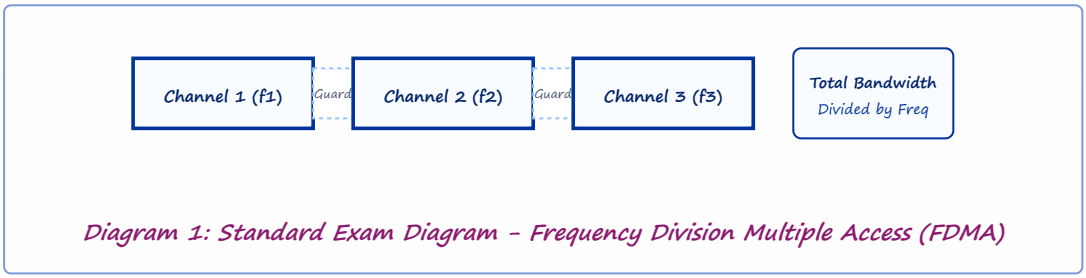


Diagram 1: Standard Exam Diagram – Frequency Division Multiple Access (FDMA)

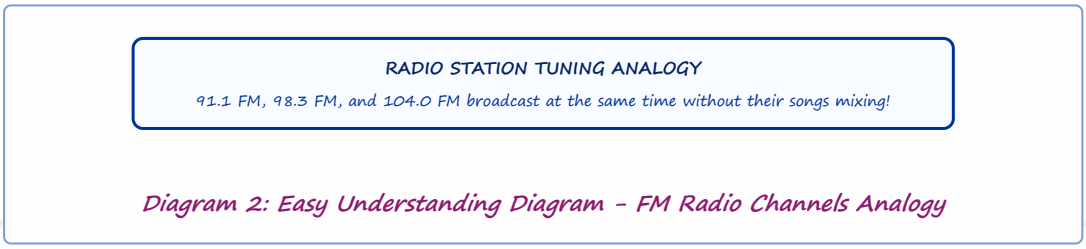


Diagram 2: Easy Understanding Diagram – FM Radio Channels Analogy

Diagram 1: Standard Exam Diagram

Shows total bandwidth divided horizontally into Frequency Channels (Channel 1, Channel 2, Channel 3) separated by Guard Bands, with time flowing continuously along the horizontal axis.

Diagram 2: Easy Understanding Diagram

Shows multiple radio towers broadcasting at 93.5 MHz, 98.3 MHz, and 104 MHz into one shared sky.

5. Advantages and Disadvantages of FDMA

Advantages	Disadvantages
Simple Technology: Uses basic analog bandpass filters; easy to implement.	**Inflexible Capacity:** If a user is not talking, their frequency band sits completely idle and wasted.
Continuous Transmission: Stations transmit continuously without taking turns.	**Guard Band Waste:** Guard bands consume valuable radio spectrum.
Zero Collisions: Frequencies are completely isolated from each other.	**Crosstalk Risk:** Cheap filters can allow adjacent channels to bleed into each other.

6. Conclusion

FDMA provides a simple, continuous transmission path by dividing frequency, making it ideal for broadcast radio and satellite uplinks, though it has been combined with TDMA and CDMA in modern cellular systems.

Q5. *(Paper-2)*

Explain pure ALOHA and Slotted ALOHA with diagrams?

1. Introduction

In the early 1970s, Norman Abramson at the University of Hawaii invented **ALOHA** to connect computers across Hawaiian islands over wireless radio. It is the grandfather of all modern random access protocols.

2. Detailed Explanation: Pure ALOHA

- **Working Principle:** A station transmits a frame whenever it has data ready.
- **Vulnerable Time:** Any frame sent during a period equal to **two frame transmission times (2 imes T_{fr})** can collide with it!
- If another station starts transmitting T_{fr} before you, its tail collides with your head.
- If another station starts transmitting T_{fr} after you, its head collides with your tail.
- **Throughput Formula:**

$S = G \cdot e^{-2G}$

- Maximum throughput occurs at G = 0.5:

$S_{max} = 0.5 \cdot e^{-1} \approx \mathbf{18.4\%}$

3. Detailed Explanation: Slotted ALOHA

- **Why It Was Invented:** To double the throughput of Pure ALOHA by cutting the vulnerable time in half!
- **Working Principle:**
- Time is divided into discrete, synchronized slots of length T_{fr}.
- A station is **not allowed to transmit whenever it wants**. It can only transmit at the **beginning of a time slot**.
- If a station misses the start of a slot, it must wait for the next slot.
- **Vulnerable Time:** Reduced to exactly **one frame time (T_{fr})**!
- **Throughput Formula:**

$S = G \cdot e^{-G}$

- Maximum throughput occurs at G = 1.0:

$S_{max} = 1.0 \cdot e^{-1} \approx \mathbf{36.8\%}$

4. Real-Life Analogy

- **Pure ALOHA is like Crossing a Highway Blindly:** You step onto the road whenever you feel like it. You are vulnerable to cars that came before you and cars coming after you. High accident rate!
- **Slotted ALOHA is like a Traffic Signal with a Crosswalk:** You are only allowed to step onto the street when the green light turns on. Accidents drop by 50%!

5. Dual Diagrams

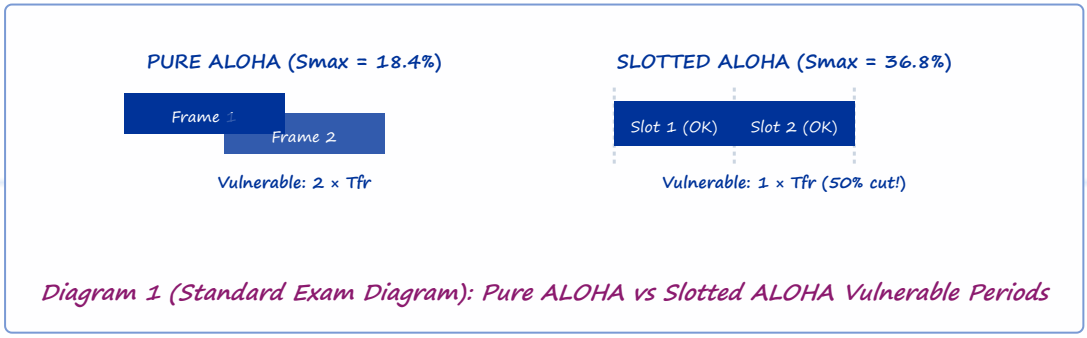


Diagram 1 (Standard Exam Diagram): Pure ALOHA vs Slotted ALOHA Vulnerable Periods

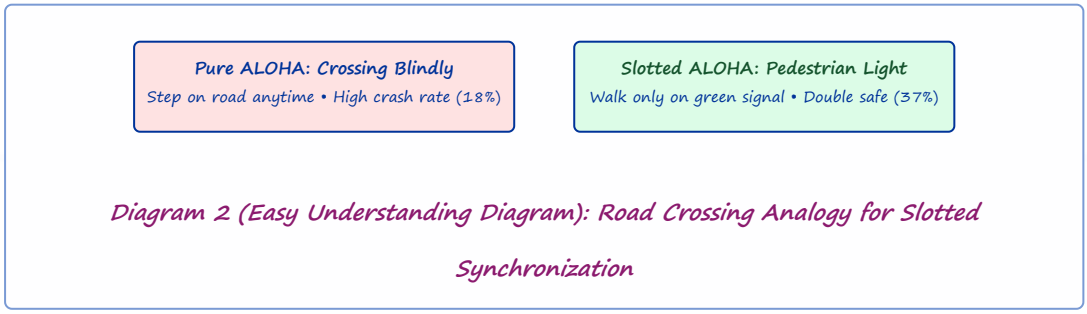


Diagram 2 (Easy Understanding Diagram): Road Crossing Analogy for Slotted Synchronization

Diagram 1: Standard Exam Diagram

Shows Pure ALOHA with random overlapping frame rectangles colliding, versus Slotted ALOHA with vertical synchronized slot lines showing collisions only when two stations pick the exact same slot.

Diagram 2: Easy Understanding Diagram

Shows Throughput (S) vs Traffic Load (G) curve comparing Pure ALOHA peak (18.4\%) with Slotted ALOHA peak (36.8\%).

6. Box Comparison: Pure ALOHA vs Slotted ALOHA

Parameter	Pure ALOHA	Slotted ALOHA
Transmission Time	Any continuous moment in time.	Only at the beginning of a discrete time slot.
Clock Synchronization	None required.	**Mandatory** master clock synchronization across all nodes.
Vulnerable Period	2 imes T_{fr}	1 imes T_{fr} (Half the vulnerable time!).
Maximum Throughput	**18.4\%** (at G = 0.5).	**36.8\%** (at G = 1.0 — Exactly double!).
Collision Probability	Very High.	Moderate (Half of Pure ALOHA).

7. Conclusion

Slotted ALOHA doubles the efficiency of random access by introducing time synchronization, laying the conceptual groundwork for modern cellular and TDMA systems.

Q6. *(Paper-2)*

List the differences between Virtual Circuit and Datagram Networks?

1. Introduction

At the boundary between the Data Link Layer and Network Layer, packet switching is handled in two fundamentally different ways: **Virtual Circuit Networks** (connection-oriented) and **Datagram Networks** (connectionless).

2. Definitions in Simple English

- **Virtual Circuit Network:** Acts like a telephone call. A dedicated logical path is set up through all routers before any data packets are sent. All packets follow the exact same path

in order.

- **Datagram Network:** Acts like sending postal letters. Each packet is treated as an independent message. Routers evaluate each packet separately and route it along whichever path is currently fastest.

3. Dual Diagrams

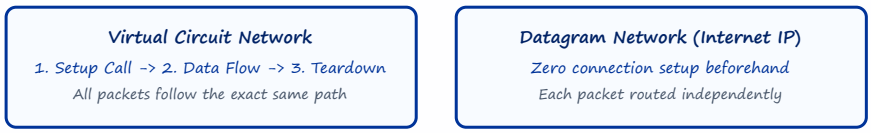


Diagram 1: Standard Exam Diagram – Virtual Circuit vs Datagram Network



Diagram 2: Easy Understanding Diagram – Railway Track vs City Streets Analogy

Diagram 1: Standard Exam Diagram

Shows Virtual Circuit path (A o R1 o R2 o B with fixed VC ID table) versus Datagram routing where Packet 1 takes top route and Packet 2 takes bottom route.

Diagram 2: Easy Understanding Diagram

Shows train on fixed railroad tracks (Virtual Circuit) versus cars driving independently through city streets (Datagram).

4. Comprehensive Box Comparison Table (8 Parameters)

S.No	Parameter	Virtual Circuit Network (e.g., ATM, Frame Relay)	Datagram Network (e.g., The Internet / IP)
1	**Connection Setup**	**Mandatory:** 3 phases (Setup, Data Transfer, Teardown).	**None:** Packets sent immediately with no setup.
2	**Packet Path**	All packets follow the **exact same pre-routed path**.	Each packet chooses its own path independently.
3	**Packet Arrival Order**	Guaranteed **strictly in order**.	Packets may arrive **out of order**; receiver reorders.

S.No	Parameter	Virtual Circuit Network (e.g., ATM, Frame Relay)	Datagram Network (e.g., The Internet / IP)
4	**Packet Header Size**	Small header (contains only a short Virtual Circuit ID — VC ID).	Large header (must contain full 32-bit/128-bit source & destination IP).
5	**Router State**	Routers maintain state tables for every open connection.	Routers are stateless; do not track connections.
6	**Impact of Router Failure**	If one router crashes, the entire connection dies.	Highly robust; routers automatically route around dead nodes.
7	**Quality of Service (QoS)**	Easy to guarantee bandwidth and delay.	Difficult; best-effort delivery only.
8	**Real-Life Example**	ATM, X.25, MPLS.	Modern Internet Protocol (IPv4 / IPv6).

5. Conclusion

Virtual circuit networks excel in predictable delay and guaranteed bandwidth for telephony, while datagram networks dominate the Internet due to their immense resilience and flexibility.

Q7. *(Paper-2)*

Explain Reservation and Polling Controlled Access methods?

1. Introduction

Controlled access methods eliminate collisions by making stations take turns. Two widely studied controlled access methods are **Reservation** and **Polling**.

2. Detailed Explanation: Reservation Method

- **How It Works:**
- Time is divided into intervals. Each interval consists of a **Reservation Frame** followed by multiple **Data Frames**.
- If there are N stations on the network, the reservation frame contains N small one-bit slots.
- If Station 1 wants to send data, it writes 1 in Slot 1.
- If Station 3 wants to send data, it writes 1 in Slot 3.

- After the reservation frame ends, Station 1 transmits its data frame, followed immediately by Station 3.
- **Benefits:** Zero collisions, 100% predictable transmission schedule.

3. Detailed Explanation: Polling Method

- **Architecture:** Uses a **Primary Station (Master)** and several **Secondary Stations (Slaves)**.
- **Operation 1 — Poll (Receiving Data):**
- The primary sends a `Poll` message to Secondary 1: *"Do you have data to send?"*
- If Secondary 1 has data, it transmits its data frame.
- If Secondary 1 has no data, it replies with a `NAK`. The primary then polls Secondary 2.
- **Operation 2 — Select (Sending Data):**
- When the primary wants to send data to Secondary 3, it sends a `Select` packet: *"Are you ready to receive?"*
- Once Secondary 3 replies `ACK`, the primary transmits the data frame.
- **Benefits:** Complete centralized control; no station can talk out of turn.

4. Real-Life Analogy

- **Reservation:** Booking a doctor's appointment online — you pick a time slot in advance, and when your time arrives, you walk in without waiting.
- **Polling:** A judge in a courtroom asking lawyers one by one: *"Does the defense have an objection? Does the prosecution have an objection?"*

5. Dual Diagrams

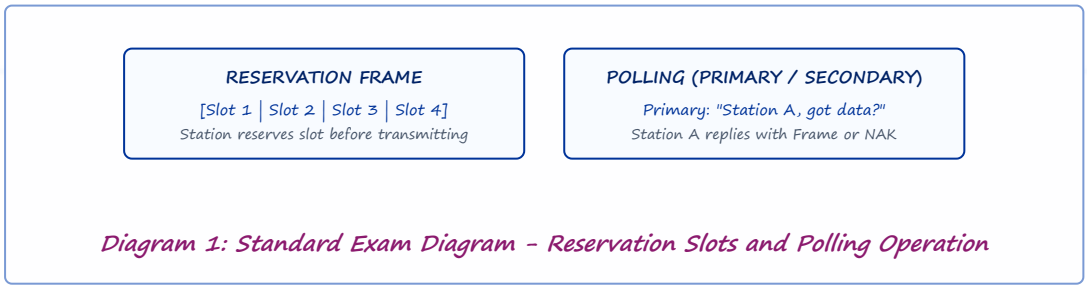


Diagram 1: Standard Exam Diagram – Reservation Slots and Polling Operation

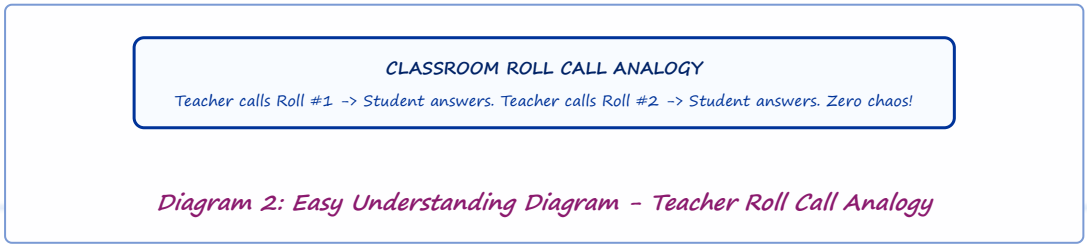


Diagram 2: Easy Understanding Diagram – Teacher Roll Call Analogy

Diagram 1: Standard Exam Diagram

Shows Reservation interval with mini-slots 1 to 4 preceding data transmissions, and Polling sequence between Primary and Secondaries.

Diagram 2: Easy Understanding Diagram

Shows airline ticket reservation seat map versus classroom teacher taking roll-call attendance.

6. Comparison Table: Reservation vs Polling

Parameter	Reservation Method	Polling Method
Control Style	Distributed (all nodes see reservation frame).	Centralized (one master computer controls all).
Overhead	Fixed mini-slot reservation frame.	Continuous polling messages (`Poll`, `Select`, `NAK`).
Failure Risk	Low.	High (if Primary crashes, whole network fails).
Efficiency	High under heavy traffic.	Low under light traffic due to polling empty stations.

7. Conclusion

Reservation allows stations to self-schedule transmissions efficiently, while Polling enforces centralized master-slave control, widely used in industrial automated manufacturing.

Q8. *(Paper-2)*

Explain different types of Time Division Multiple Access (TDMA)?

1. Introduction

In digital communications, dividing a channel by frequency (FDMA) wastes spectrum when users are silent. **TDMA (Time Division Multiple Access)** solves this by giving users the **entire channel bandwidth**, but dividing time into discrete periodic time slots.

2. Definition & Core Working

- **TDMA:** A channelization technique where the entire frequency band is allocated to one user at a time for a brief, periodic slice of time.
- Time is organized into repeating **Frames**, and each frame is divided into N **Time Slots**.
- Station 1 transmits during Slot 1, Station 2 during Slot 2, and so on.

- **Guard Times:** Small buffer gaps placed between consecutive time slots to ensure clock drift does not cause one station's burst to collide with another.

3. Major Types of TDMA

A. Synchronous (Fixed) TDMA

- Each station is assigned a fixed, permanently dedicated time slot in every frame.
- If a station has no data to send, its time slot remains **completely empty and wasted**.
- Used in 2G GSM cellular networks.

B. Asynchronous (Statistical) TDMA

- Time slots are not pre-assigned. Instead, slots are dynamically allocated **on demand** only to stations that currently have active data to transmit.
- Eliminates empty wasted slots, dramatically increasing channel efficiency.

4. Dual Diagrams

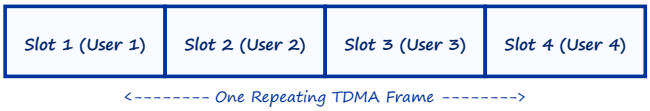


Diagram 1: Standard Exam Diagram – Time Division Multiple Access (TDMA) Slots



Diagram 2: Easy Understanding Diagram – Scheduled Appointment Slots Analogy

Diagram 1: Standard Exam Diagram

Shows repeating TDMA Frame divided into Slot 1, Slot 2, Slot 3 separated by Guard Times, all operating across the full frequency spectrum.

Diagram 2: Easy Understanding Diagram

Shows revolving door or turnstile at a train station — only one commuter passes through each turn of the door.

5. Comparison: FDMA vs TDMA

Parameter	FDMA	TDMA
Division Axis	Divided by **Frequency**.	Divided by **Time**.

Parameter	FDMA	TDMA
Bandwidth Used	Slices of bandwidth used continuously.	Full bandwidth used in bursts.
Synchronization	No time sync needed.	**Strict time synchronization mandatory**.
Guard Separation	Separated by Guard Bands in frequency.	Separated by Guard Times in time.
Cellular Standard	1G AMPS.	2G GSM.

6. Conclusion

TDMA maximized spectrum utilization in 2G digital cellular systems by digitizing voice into small bursts, eliminating the need for bulky analog frequency filters.

Q9. *(Paper-3)*

Explain reservation-based MAC with an example.

1. Introduction & Concept

When multiple stations share a high-bandwidth satellite or fiber channel, letting them transmit randomly causes massive collisions. **Reservation-Based MAC** eliminates collisions by forcing stations to declare their intention to transmit in advance during a small reservation period.

2. Step-by-Step Working Mechanism

- Time is divided into synchronized intervals.
- Each interval begins with a **Reservation Frame** followed by M **Data Frames**.
- If there are N stations in the network, the reservation frame contains N mini-slots:

- Station 1 owns Mini-slot 1.
- Station 2 owns Mini-slot 2.
- Station N owns Mini-slot N.

- **Reservation Step:** Any station with data to send marks its mini-slot with a 1. Stations with no data leave it 0.
- **Transmission Step:** All stations observe the reservation frame. They transmit their data frames in order:

- If Stations 1 and 4 marked a 1, Station 1 transmits its data frame first, followed immediately by Station 4.

3. Real-Life Example

Imagine a university seminar where 5 students want to present research:

- At 9:00 AM, the professor asks: "Raise your hand if you have a presentation today" (Reservation Mini-Slots).
- Students A and D raise their hands.
- Student A presents from 9:05 to 9:30 AM.
- Student D presents from 9:30 to 10:00 AM.
- Zero collisions occur because everyone knows exactly whose turn it is.

4. Dual Diagrams

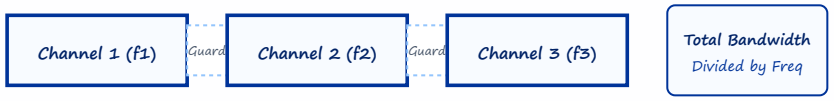


Diagram 1: Standard Exam Diagram – Frequency Division Multiple Access (FDMA)



Diagram 2: Easy Understanding Diagram – FM Radio Channels Analogy

Diagram 1: Standard Exam Diagram

Shows Interval k with 4 mini-slots (1 0 1 0), followed by Frame 1 and Frame 3 data transmissions.

Diagram 2: Easy Understanding Diagram

Shows sign-up sheet on a conference room door where speakers write their names before using the room.

5. Advantages and Disadvantages

Advantages	Disadvantages
Zero Collisions: Guaranteed clean channel during data transfer.	Reservation frame mini-slots consume bandwidth extra check bits.

Advantages	Disadvantages
High Efficiency at Heavy Load: Channel is 100% busy when demand is high.	High latency under light load (must wait for reservation frame).

6. Conclusion

Reservation-based MAC provides an orderly, collision-free schedule that is widely used in satellite communication systems where collision retransmissions would cause intolerable delays.

Q10. *(Paper-3)*

Explain FDMA and TDMA.

1. Introduction

The two classic channelization techniques used to share electromagnetic spectrum among multiple concurrent users are **FDMA** (Frequency Division) and **TDMA** (Time Division).

2. FDMA (Frequency Division Multiple Access)

- **Principle:** Slices total frequency bandwidth into separate sub-channels.
- Each station is tuned to a dedicated frequency band and transmits **continuously**.
- Uses **Guard Bands** to avoid spectral overlap.
- *Example:* Radio broadcasting stations (FM radio).

3. TDMA (Time Division Multiple Access)

- **Principle:** Allocates the entire frequency channel to one user at a time for a **brief periodic time slot**.
- Stations take turns transmitting high-speed bursts.
- Uses **Guard Times** to absorb clock jitter.
- *Example:* 2G GSM cellular networks.

4. Dual Diagrams

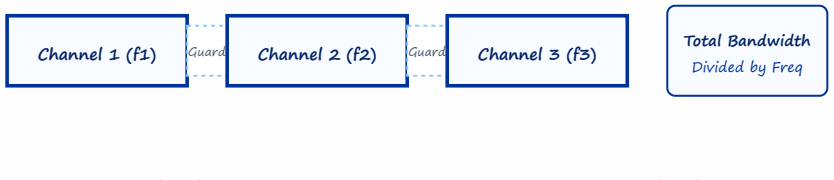


Diagram 1: Standard Exam Diagram - Frequency Division Multiple Access (FDMA)



Diagram 2: Easy Understanding Diagram – FM Radio Channels Analogy

Diagram 1: Standard Exam Diagram

Side-by-side technical diagrams contrasting FDMA (horizontal frequency channels separated by guard bands) with TDMA (vertical time slots separated by guard times).

Diagram 2: Easy Understanding Diagram

Shows FDMA as a 3-lane swimming pool where swimmers stay in their own lanes, versus TDMA as a single diving board where divers take turns one after another.

5. Comprehensive Comparison: FDMA vs TDMA

Parameter	FDMA	TDMA
Resource Divided	**Frequency spectrum** is divided.	**Time axis** is divided.
User Signal	Continuous analog or digital signal.	Discontinuous high-speed digital bursts.
Synchronization	No time sync needed.	**Strict microsecond time sync mandatory**.
Buffer Requirement	No buffering needed at transmitter.	Must buffer voice data and blast in short bursts.
Guard Protection	**Guard Bands** in frequency domain.	**Guard Times** in time domain.

6. Conclusion

FDMA divides space by frequency, while TDMA divides space by time. Modern wireless systems (like 4G LTE and 5G) combine both techniques into mutually independent (orthogonal) frequency-division multiple access (OFDMA).

Q11. *(Paper-3)*

Explain Standard Ethernet and its characteristics.

1. Introduction

Standard Ethernet was standardized under **IEEE 802.3** in 1985. It established the core principles of Local Area Networks (LANs) that remain in use today across homes, universities, and enterprise data centers.

2. Key Characteristics of Standard Ethernet

- **Transmission Speed: 10 Mbps** (Megabits per second).
- **Media Access Protocol: CSMA/CD** with 1-persistent carrier sensing and binary exponential backoff.
- **Network Topologies:**
- **Bus Topology:** Using thick coaxial (10BASE5) or thin coaxial (10BASE2).
- **Star Topology:** Using twisted-pair copper cables connected to a central hub (10BASE-T).
- **Physical Addressing:** Uses unique, permanent **48-bit MAC Addresses** burned into each network interface card.

3. Structure of the Standard IEEE 802.3 Ethernet Frame

An Ethernet frame has seven distinct fields:

- **Preamble (7 Bytes):** Alternating `10101010` pattern to synchronize receiver clock pulses.
- **SFD (Start Frame Delimiter — 1 Byte):** `10101011` signaling that the frame begins next.
- **Destination MAC Address (6 Bytes):** Hardware address of the destination device.
- **Source MAC Address (6 Bytes):** Hardware address of the sending device.
- **Length / Type (2 Bytes):** Specifies payload length or protocol type (e.g., IPv4 = `0x0800`).
- **Data Payload (46 to 1500 Bytes):** Actual network packet. (If data is < 46 bytes, padding is added to reach the 64-byte minimum frame size).
- **FCS (Frame Check Sequence — 4 Bytes):** 32-bit CRC error detection code.

4. The 64-Byte Minimum Frame Rule (Why It Matters)

- Minimum frame size is strictly **64 Bytes (512 bits)**.
- Transmission time for 512 bits at 10 Mbps is:

$$T_{tx} = \frac{512 \text{ bits}}{10 \text{ Mbps}} = 51.2\ \mu\text{s}$$

- This time is exactly equal to the maximum round-trip propagation time (2 au) across a 2500-meter network. If a collision happens at the furthest corner, the echo reaches the sender before it finishes sending 64 bytes, ensuring collisions are always detected!

5. Dual Diagrams

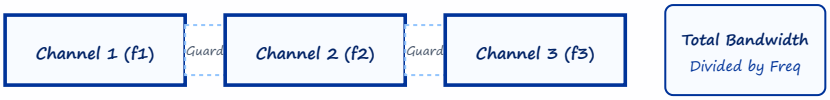


Diagram 1: Standard Exam Diagram – Frequency Division Multiple Access (FDMA)



Diagram 2: Easy Understanding Diagram – FM Radio Channels Analogy

Diagram 1: Standard Exam Diagram

Shows the complete 802.3 Ethernet frame layout with byte counts for all 7 fields.

Diagram 2: Easy Understanding Diagram

Shows an envelope with Preamble stamp, Sender/Receiver addresses, letter payload, and wax seal CRC at the end.

6. Summary Box: Standard Ethernet Specifications

Feature	Specification
Data Rate	10 Mbps
Access Method	CSMA/CD
Frame Size Range	64 Bytes minimum to 1518 Bytes maximum
Address Length	48 bits (6 Bytes) • Hex format: `00:1A:2B:3C:4D:5E`
Error Check	CRC-32 (4-byte FCS)

7. Conclusion

Standard Ethernet's elegant framing, unique MAC addressing, and robust collision handling established the blueprint for modern digital local networking.

Q12. *(Paper-3)*

Explain working of token passing with example.

1. Introduction

In contention protocols, collisions waste bandwidth when traffic is heavy. **Token Passing** is a controlled access method that eliminates collisions completely by making computers take turns using a circulating permission token.

2. Definition & Core Working

- **Token:** A special, compact frame that circulates continuously around a logical ring of computers.
- **Rules of the Protocol:**
  - If a computer has no data to send, it simply passes the token to its neighbor.
  - If a computer has data to send, it **captures the token**.
  - The computer changes the token status from "free" to "busy", attaches its data frame, and sends it down the ring.
  - While the computer holds the token, **no other computer is allowed to transmit**.
  - The destination computer reads the data, marks the frame as "acknowledged", and passes it along.
  - When the frame returns to the sender, the sender strips the data and releases a **free token** for the next station.

3. Real-Life Analogy

Imagine a meeting in a conference room with a **Single Microphone**:

- Only the person holding the microphone is permitted to speak.
- When they finish speaking, they hand the microphone to the person sitting to their right.
- No two people ever speak at once, so zero arguments (collisions) occur.

4. Dual Diagrams

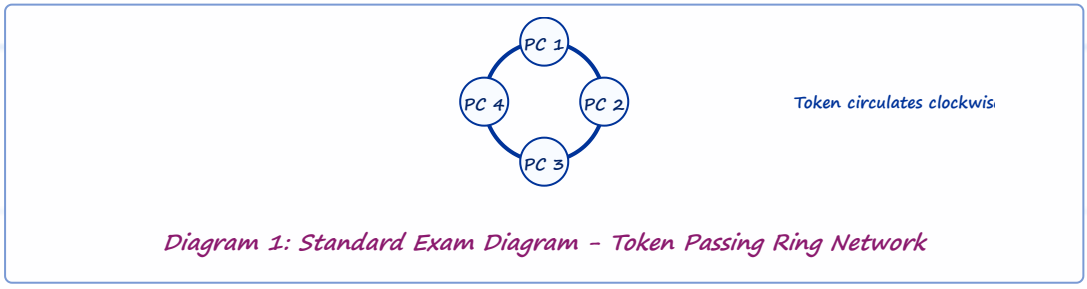


Diagram 1: Standard Exam Diagram – Token Passing Ring Network

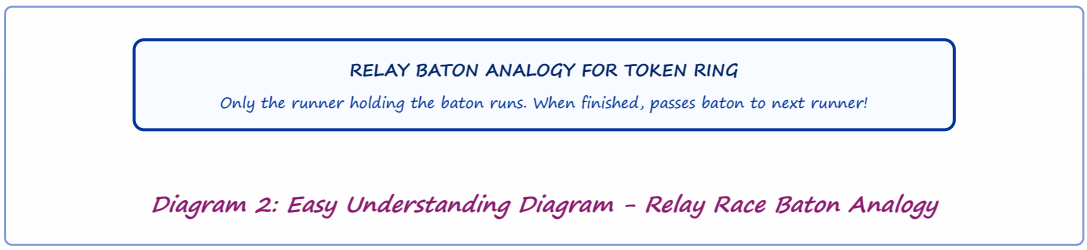


Diagram 2: Easy Understanding Diagram – Relay Race Baton Analogy

Diagram 1: Standard Exam Diagram

Shows 4 stations in a ring (A, B, C, D) showing Token capture, Data frame circulation to Station C, and Free Token regeneration.

Diagram 2: Easy Understanding Diagram

Shows children sitting in a circle passing a talking stick or toy baton.

5. Advantages and Disadvantages of Token Passing

Advantages	Disadvantages
Zero Collisions: Guaranteed collision-free operation at all times.	**Token Loss Vulnerability:** If a station crashes while holding the token, the token vanishes and network stops.
Fairness: Every computer gets an equal turn; no computer starves.	**High Latency at Low Load:** Even if only 1 computer wants to talk, it must wait for the token to circle around.
Predictable Delivery: Guaranteed maximum waiting time, ideal for factories.	Complex ring maintenance protocols required (token monitors).

6. Conclusion

Token passing provides deterministic, collision-free performance, making it the preferred choice for industrial automation and high-reliability factory floor networks (IEEE 802.5 / FDDI).

Q13. *(MID-1)*

Explain channelization techniques: FDMA, TDMA and CDMA.

1. Introduction

When multiple wireless transmitters share the same physical airspace, they must be separated so their signals do not scramble each other. The three primary channelization technologies that power modern cellular networks are **FDMA**, **TDMA**, and **CDMA**.

2. Detailed Breakdown of the Three Techniques

A. FDMA (Frequency Division Multiple Access)

- **Concept:** Slices frequency bandwidth into separate channels.
- Users transmit simultaneously on **different frequencies**.
- Analogy: Different radio stations broadcasting on different FM frequencies.

- Generation: **1G Analog Cellular** (AMPS).

B. TDMA (Time Division Multiple Access)

- **Concept:** Slices time into periodic time slots.
- Users transmit sequentially on the **same frequency**, but at **different times**.
- Analogy: People taking turns speaking through a walkie-talkie.
- Generation: **2G Digital Cellular** (GSM).

C. CDMA (Code Division Multiple Access)

- **Concept:** All users transmit on the **same frequency at the exact same time**!
- Each user's data is multiplied by a unique mathematical **Orthogonal Chip Code** (Walsh Code).
- The receiver takes the mixed radio signals and multiplies by the sender's code; because the codes are mutually independent (orthogonal), all other users' signals sum to zero, leaving only the desired message!
- Analogy: A room where one pair speaks English, another speaks French, and another speaks Japanese. Even though everyone speaks at once, you only listen to and understand your own language!
- Generation: **3G Cellular** (IS-95, UMTS).

3. Step-by-Step Numerical Example: CDMA Orthogonality

Let Station 1 have Code $C_1 = [+1, +1]$ and Station 2 have Code $C_2 = [+1, -1]$.

- Check mutually independent (orthogonal)ity: Inner dot product:

$C_1 \cdot C_2 = (+1)(+1) + (+1)(-1) = 1 - 1 = \mathbf{0}$

- Suppose Station 1 sends bit $d_1 = +1$ and Station 2 sends bit $d_2 = -1$:
- Signal 1: $S_1 = d_1 imes C_1 = +1 imes [+1, +1] = [+1, +1]$
- Signal 2: $S_2 = d_2 imes C_2 = -1 imes [+1, -1] = [-1, +1]$
- Combined air signal:

$S = S_1 + S_2 = [+1 - 1,\ +1 + 1] = [0, +2]$

- **Receiver extracting Station 1's bit:**

$d_1 = \rac{S \cdot C_1}{2} = \rac{(0)(+1) + (+2)(+1)}{2} = \rac{2}{2} = \mathbf{+1} ext{ (Perfect Recovery!)}$

4. Dual Diagrams

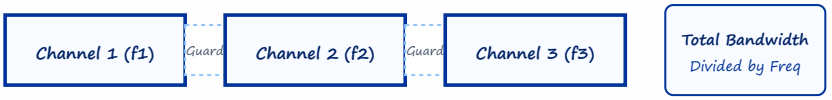


Diagram 1: Standard Exam Diagram – Frequency Division Multiple Access (FDMA)



Diagram 2: Easy Understanding Diagram – FM Radio Channels Analogy

Diagram 1: Standard Exam Diagram

Shows 3D coordinate box comparing FDMA (Frequency slices), TDMA (Time slices), and CDMA (Code dimensions).

Diagram 2: Easy Understanding Diagram

Shows 3 pairs of people in a room speaking English, Spanish, and French simultaneously without interference.

5. Master Comparison Table: FDMA vs TDMA vs CDMA

Parameter	FDMA	TDMA	CDMA
How It Divides	Divides **Frequency**.	Divides **Time**.	Divides **Codes**.
Transmission Time	Continuous.	Bursts during slot.	Continuous.
Transmission Frequency	Dedicated sub-band.	Full band.	Full band.
Synchronization	None needed.	Strict time sync.	Complex chip sync.
Cellular Era	1G	2G	3G
Security & Privacy	Low.	Moderate.	**Extremely High** (Spread Spectrum).

6. Conclusion

Channelization evolved from simple frequency slicing (FDMA) to digital time sharing (TDMA) to sophisticated mathematical coding (CDMA), continuously multiplying the capacity of

cellular communications.

Q14. *(Paper-3)*

Explain the design issues in the network layer.

1. Introduction

The **Network Layer (Layer 3)** is the heart of the Internet. Its primary job is to deliver packets across multiple network hops from the original source host to the final destination host. Designing a reliable network layer involves solving four fundamental engineering challenges.

2. Four Major Design Issues in the Network Layer

A. Routing (Finding the Best Path)

- When a packet must travel from India to the United States, hundreds of alternative paths exist through satellites, undersea cables, and intermediate routers.
- The routing algorithm must evaluate line speed, cost, and delays to calculate the shortest and fastest route (using algorithms like Dijkstra or Bellman-Ford).

B. Congestion Control (Preventing Traffic Jams)

- If too many computers pump packets into a router faster than the router can forward them, the router's internal queue overflows and packets drop.
- The network layer must detect congestion early and throttle sender speeds using algorithms like Leaky Bucket and Token Bucket.

C. Quality of Service (QoS) & Delay Guarantees

- Different applications have different needs:
- Email doesn't care if a letter is delayed by 3 seconds.
- Live voice calls (VoIP) and Zoom video meetings break down if packets are delayed by more than 150 milliseconds.
- The network layer must prioritize real-time traffic over bulk file downloads.

D. Internetworking (Connecting Incompatible Networks)

- A packet may leave an Ethernet wired LAN, cross an ATM fiber backbone, pass through a wireless satellite link, and arrive on a mobile 5G smartphone.
- Each network has different packet size limits (MTU), addressing schemes, and protocols. The network layer must handle **Packet Fragmentation and Reassembly** to bridge incompatible networks.

3. Real-Life Analogy

Think of the **National Postal & Highway System**:

- **Routing:** Google Maps picking the best highway to avoid closed roads.
- **Congestion Control:** Metering lights at highway on-ramps to prevent traffic gridlock.
- **QoS:** Giving priority sirens to ambulances over regular delivery trucks.
- **Internetworking:** Transferring cargo containers smoothly from cargo ships to freight trains to local vans.

4. Dual Diagrams

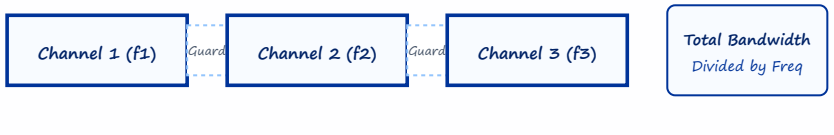


Diagram 1: Standard Exam Diagram – Frequency Division Multiple Access (FDMA)



Diagram 2: Easy Understanding Diagram – FM Radio Channels Analogy

Diagram 1: Standard Exam Diagram

Shows the 4 design pillars of the Network Layer: Routing Engine, Congestion Monitor, QoS Scheduler, and Internetworking Gateway.

Diagram 2: Easy Understanding Diagram

Shows road traffic management: Highway signs (Routing), Traffic lights (Congestion), Ambulance lane (QoS), and Train-to-Truck transfer depot (Internetworking).

5. Summary Box: Network Layer Design Challenges

Design Challenge	Core Goal in Simple English	Main Technique / Algorithm
Routing	Find shortest, fastest path.	Dijkstra, OSPF, Distance Vector.
Congestion Control	Prevent router queue overflows.	Leaky Bucket, Token Bucket, Choke Packets.

Design Challenge	Core Goal in Simple English	Main Technique / Algorithm
Quality of Service	Guarantee bandwidth for video/voice.	Traffic Shaping, Priority Queuing.
Internetworking	Connect diverse hardware networks.	Fragmentation, Reassembly, Gateways.

6. Conclusion

Solving routing, congestion, quality of service, and internetworking allows the Network Layer to unite thousands of independent, diverse networks into one single global Internet.